

A Simple Treatment of Physical Nonequilibrium Water Flow in Soils

P. J. Ross and K. R. J. Smettem*

ABSTRACT

Water flow in soils with sufficiently large aggregate size or pore-class heterogeneity may exhibit a nonequilibrium between the actual water content and that given by the water retention curve. The result is deeper penetration of infiltrating water than predicted using classical infiltration theory. Models of this process usually divide the soil into two or more exchanging flow regions. A simpler treatment is possible by combining Richards' equation with a dynamic description of the approach to equilibrium. We present a first-order time constant equilibration model of infiltration into a hypothetical structured soil and use the model to describe published outflow responses to constant rate rainfall on six large, undisturbed cores. Using measured hydraulic properties and varying only the A and B horizon time constants, the model was fitted to cumulative outflow from one particular soil core that had measured time domain reflectometry (TDR) water contents at the 0.05- and 0.5-m depths recorded during the experiment. Cumulative outflow was fitted using time constants of 4 and 5 h for A and B horizons, respectively, and this also gave good agreement with TDR measured water contents. Cumulative outflow and runoff from a further four of the six cores was described using the same A and B horizon time constants and varying only the macropore hydraulic conductivity. The remaining core contained a decayed root, which conducted water rapidly with little opportunity for lateral exchange. A description of cumulative outflow required both the macropore hydraulic conductivity and the time constant to be altered.

SOIL PHYSICAL DESCRIPTIONS of water flow in soils have traditionally assumed local equilibrium at the elementary volume scale. However, in recent years, it has become clear that this assumption is inadequate for describing water flow through at least some heterogeneous field soils (Parker and Valocchi, 1986), particularly those containing flow paths around relatively large structural units (Bouma, 1981; Ogden et al., 1992). The problem lies in choosing a convenient scale for description of the process. In principle, if detailed knowledge of properties and structure were available, the system could perhaps be analyzed with the equilibrium assumption, but in practice it has not proved to be possible.

The usual response to this physical nonequilibrium situation is to classify the system into two or more zones that exchange material with one another while maintaining local equilibrium internally, an approach going some way towards the practically impossible detailed analysis. Thus Philip (1968), in an analytical investigation of water absorption in aggregated media, assumed a macroporosity, treated at a scale large compared with characteristic macropore dimensions, and a microporosity, treated as a distributed source or sink. Elaborations

on this basic approach, usually relying on numerical methods of analysis, continue to appear (e.g., Gerke and van Genuchten, 1993). Such analyses require that the separate zones and at least some of their hydraulic properties be specified, when in practice they may not be available. In this study we take a simpler, slightly different view of water flow in soils exhibiting physical nonequilibrium and suggest a simple macroscopic model requiring only a single parameter in addition to the soil water retention and hydraulic conductivity functions. We present an illustrative example using this model and then apply it to describe experimental results on large undisturbed cores of a structured field soil.

THEORY

Assume that water movement can be adequately described by the one-dimensional Richards equation

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(h) \left(\frac{\partial h}{\partial z} - 1 \right) \right] \quad [1]$$

where θ is volumetric water content, h is matric head, $K(h)$ is the soil hydraulic conductivity function, t is time, and z is depth. It is also necessary to assume the existence of an appropriate scale, large compared with the scale of structural features responsible for nonequilibrium in the horizontal plane but small compared with the overall flow region, for the definition of the variables. With the assumption of local equilibrium, $\theta = \theta_e(h)$, where $\theta_e(h)$ is the soil water retention function. Without this assumption θ becomes independent of h and some other way of specifying $\partial \theta / \partial t$ must be found. We assume that $\partial \theta / \partial t$ can be specified through the differential equation $\partial \theta / \partial t = f(\theta, \theta_e)$ where $f(\theta, \theta_e)$ is a known function of the actual and equilibrium water contents. Note that similar equations can result from a dual porosity model with flow but no water storage in one region while the other region acts only as a source or sink for water. The interpretation differs, however.

Many numerical methods for the solution of Richards' equation use discrete times $t^1, t^2, \dots, t^k, \dots$ and approximate the term $\partial \theta / \partial t$ by a backward difference. Straightforward application of this method gives $(\theta^{k+1} - \theta^k) / \Delta t \approx f(\theta^{k+1}, \theta_e^{k+1})$, where $\Delta t = (t^{k+1} - t^k)$, allowing a solution for θ^{k+1} . This solution is accurate if Δt is small enough, but for the problem considered here it is possible to obtain a better solution accurate for larger Δt by writing $\partial \theta / \partial t \approx f(\theta, \theta_e^{k+1})$ and solving to get

$$\Delta t = \int_{\theta^k}^{\theta^{k+1}} \frac{d\theta}{f(\theta, \theta_e^{k+1})} \quad [2]$$

from which to obtain θ^{k+1} . A modification to compute θ^{k+1} in this way, rather than by equating it to θ_e^{k+1} , should require little change to most numerical flow codes for solving the one-dimensional Richards equation.

We henceforth assume that $f(\theta, \theta_e^{k+1}) = (\theta_e - \theta) / \tau$, where τ is an equilibration time constant. This is one of the simplest models for the approach to equilibrium and is the usual starting point where detailed information is unavailable. From Eq.

P.J. Ross, CSIRO Land and Water, Cunningham Laboratory, 306 Carmody Road, St Lucia, Brisbane, QLD 4067, Australia; K.R.J. Smettem, Soil Science and Plant Nutrition, The University of Western Australia, Nedlands, Perth, WA. 6907, Australia. Received 16 Sept. 1999. *Corresponding author (hydsol@cyllene.uwa.edu.au).

[2], an accurate approximation is

$$\theta^{k+1} = \theta^k + (\theta_e^{k+1} - \theta^k)[1 - \exp(-\Delta t/\tau)] \quad [3]$$

Note that, for $\tau = 0$, Eq. [3] reduces to $\theta^{k+1} = \theta_e^{k+1}$. Where θ can be partitioned to correspond with a number of regions with different time constants, we simply add the rates of change. For example, if $\theta = \theta_1 + \theta_2$, where θ_1 is the nonequilibrium component and θ_2 is always at equilibrium with θ_{2e} and $\partial\theta/\partial t = (\theta_{1e} - \theta_1)/\tau + \partial\theta_2/\partial t$, where θ_{1e} is the equilibrium value for Component 1, then $\theta^{k+1} = \theta_1^k + (\theta_{1e}^{k+1} - \theta_1^k)[1 - \exp(-\Delta t/\tau)] + \theta_{2e}^{k+1}$.

MATERIALS AND METHODS

Illustrative Example

The example was for a hypothetical soil constructed by adding an exponential $K(h)$ function, representing a structural component responsible for nonequilibrium, to the hydraulic conductivity of Yolo light clay (fine-silty, mixed, nonacid, thermic Typic Xerorthent). The addition of a simple exponential $K(h)$ function to describe the macropore contribution to hydraulic conductivity was chosen for this illustration because it has been widely used in the literature (Reynolds and Elrick, 1991; Ross and Smettem, 1993; Mohanty et al., 1997). In this example we do not add any macroporosity to the water retention curve.

Adding the exponential $K(h)$ function to the parameters of Fuentes et al. (1992) for Yolo light clay, we have

$$K = K_1 \left(\frac{\theta - \theta_r}{\theta_s - \theta_r} \right)^\eta + K_2 \exp\left(-\frac{h}{h_{ek}}\right) \quad [4]$$

where θ_s is the saturated water content, θ_r is the residual water content, K is the hydraulic conductivity, K_1 is the matrix saturated hydraulic conductivity, and K_2 is the macropore saturated hydraulic conductivity, η is the conductivity exponent, and $-1/h_{ek}$ is the slope of the macropore $\ln K(h)$ (Smettem et al., 1991).

Also,

$$\frac{\theta - \theta_r}{\theta_s - \theta_r} = \left[1 + \left(\frac{h}{h_g} \right)^n \right]^{-m} \quad [5]$$

where h_g is a pressure scale parameter, while n and m are shape parameters. Note that θ in these hydraulic property descriptions corresponds with θ_e in our model. Fuentes et al. (1992) give $\theta_s = 0.495 \text{ cm}^3 \text{ cm}^{-3}$, $\theta_r = 0 \text{ cm}^3 \text{ cm}^{-3}$, $K_1 = 0.443 \text{ mm h}^{-1}$, $h_g = -193.1 \text{ mm}$, $m = 0.0995$, $n = 2/(1 - m)$, $\eta = 9.1432$. We add the hypothetical values $K_2 = 50 \text{ mm h}^{-1}$ and $h_{ek} = -50 \text{ mm}$ to the second term in the equation for conductivity. The equilibration model with a time constant τ was assumed. There were three infiltration runs with $\tau = 0 \text{ h}$ (local equilibrium), $\tau = 1 \text{ h}$, and $\tau = 2 \text{ h}$. The initial profile was at a matric head of -15000 cm , and the applied head at the surface was 0 cm . Similarly, there were three runs for drainage from saturation with a unit hydraulic gradient at the 50-cm depth. The equations were solved numerically using central differences with an inverse sinh transform at a fixed 1-cm grid spacing, and backward differences in time, as described by Ross (1990). Only very slight modification of existing code was required to incorporate the exponential equilibration.

Structured Field Soil

We applied the nonequilibrium model to the experiment of Smettem et al. (1994), which reports cumulative outflow from initially dry undisturbed soil cores subjected to artificial rainfall from a low energy sprinkler system at a constant rate

of 48 mm h^{-1} . A feature of these experiments was that, in all cases, outflow commenced prior to complete wetting of the soil cores. This phenomenon is commonly referred to as preferential flow or bypassing (Smettem et al., 1983). For these simulations, the lower boundary condition was atmospheric pressure.

Six soil cores were obtained from a long-term pasture and the soil had a distinct textural contrast at the 0.2-m depth. The surface 0.2 m was a fine sandy loam and the subsurface a predominantly kaolinitic clay. Cores were 0.3 m in diameter and between 0.63 and 0.67 m long. To identify individual cores, we have retained the numbering scheme (1–6) used by Smettem et al. (1994).

In situ saturated and near-saturated hydraulic conductivities for this soil were reported by Smettem et al. (1991) and successfully used to model field drainage by Richards and Smettem (1992). Smettem et al. (1994) also reported values of macroporosity θ_{mp} for the large undisturbed cores. For the present study, water retention data was obtained from undisturbed soil cores of 7.5-cm diameter and 5.0-cm length. Six cores were taken from two depths ($0.05\text{--}0.1$ and $0.3\text{--}0.35 \text{ m}$ depths) at the location from which the large undisturbed cores were extracted. Water retention measurements were made using hanging water columns at -0.1 , -0.3 , -0.5 , and $-1.0 \text{ m H}_2\text{O}$. Subsamples were then equilibrated on pressure plates at -10 and $-150 \text{ m H}_2\text{O}$. The resulting data characterizes the soil texture and was fitted using the nonlinear least square retentention curve fitting program RETC (van Genuchten et al., 1991), with the water retention model given by Eq. [5], but with θ_s replaced by the saturated water content of the soil matrix $\theta_{mx} = \theta_s - \theta_{mp}$. The full water retention equation used was given by Ross and Smettem (1993) and adds an exponential term to the van Genuchten water retention curve in order to account for macroporosity as follows

$$\theta - \theta_r = (\theta_{mx} - \theta_r) \left[1 + \left(\frac{h}{h_b} \right)^n \right]^{-m} + \theta_{mp} \left(1 + \frac{h}{h_e} \right) \exp\left(-\frac{h}{h_e}\right) \quad [6]$$

From the data of Smettem et al. (1991, 1994) we took θ_{mp} as $0.05 \text{ m}^3 \text{ m}^{-3}$ and h_e (macropore pressure scale parameter) as -10 mm for the $0\text{--}20\text{-cm}$ layer and -20 mm for depths $>20 \text{ cm}$.

The conductivity was given by Eq. [4] and it was assumed that macropore water content was in equilibrium with the matric potential. We used the relations $m = 1 - 2/n$ and $\eta = 1/m + 2$ (Fuentes et al., 1992).

Initial modeling focused on Core 4 of Smettem et al. (1994) because this core had additional unpublished TDR water content data available for depths of 0.05 and 0.5 m . The horizontally inserted TDR probes, of 0.15-m length, were assumed to measure water content for a sufficient volume to average across the structural features responsible for the nonequilibrium flow through the core.

The hydraulic properties used in the flow simulations are shown in Table 1, which also indicates which parameters were obtained from previously published data, which were obtained from water retention measurements, and which were adjusted in the model runs. The only parameter initially calibrated for Core 4 by fitting the model to the cumulative drainage data was τ . The fitting was performed by altering the value of τ independently for the A and B horizons until the best match to the experimental data was obtained. The nonequilibrium model was then used to generate water content changes at 0.05 and 0.5 m within the core for comparison with the experimental TDR data.

Table 1. Parameters used in the time constant modelling of drainage from Cores 4, 3, and 1 of Smettem et al. (1994).

Core	Depth	Parameter									
		θ_{mx}	θ_r	θ_{mp}	m	h_g	h_c	h_{ck}	K_1	K_2	τ
	m	$m^3 m^{-3}$			$(1 - 2/n)$	mm			mm h ⁻¹		h ⁻¹
4	0–0.2	0.40 [†]	0 [‡]	0.05 [‡]	0.09 [‡]	–52 [‡]	–10 [†]	–40 [†]	14 [†]	230 [†]	4.0 [§]
4	0.2–0.63	0.40 [†]	0 [‡]	0.05 [‡]	0.055 [‡]	–180 [‡]	–20 [†]	–90 [†]	5 [†]	26 [†]	5.0
3	0–0.2	0.40	0	0.05	0.09	–52	–10	–40	14	22	4.0
3	0.2–0.63	0.40	0	0.05	0.055	–180	–20	–90	5	11	5.0
1	0–0.2	0.40	0	0.05	0.09	–52	–10	–40	2.5	2000	9.2
1	0.2–0.67	0.40	0	0.05	0.055	–52	–10	–90	5	2000	9.2

[†] Parameters from Smettem et al. (1991).

[‡] Parameters from undisturbed small cores (this study).

[§] Parameters varied in a particular simulation are underlined.

Subsequent to the initial analysis, we also investigated the ability of the model to describe the remaining five cores reported by Smettem et al. (1994). Core 3 had the least cumulative drainage across 6 h and had 45 mm of surface runoff. It was necessary to fit values of K_2 (Eq. [4]), but the same values of τ as for Core 4 were satisfactory. By simply varying K_2 we were able to describe the drainage curves for five of the six cores reported by Smettem et al. (1994). However, the remaining core (Core 1) was traversed by a decaying root and had 230 mm of through drainage and no surface runoff. It was necessary to fit values of K_1 (horizon A), K_2 , and τ .

RESULTS AND DISCUSSION

Illustrative Example

Water content profiles for the infiltration runs are shown in Fig. 1. The equilibration time constant had a major effect on the wetting profile, particularly at early times. The depth of penetration was initially much less with local equilibrium, and the water content was much

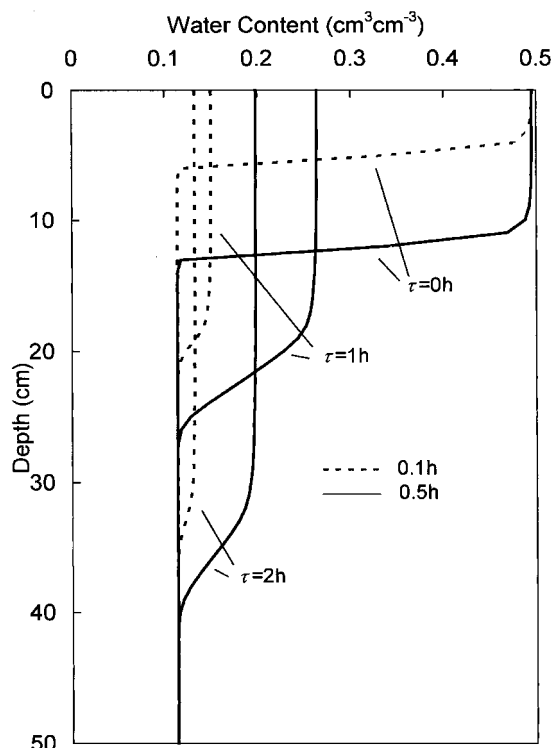


Fig. 1. Water content profiles at 0.1 and 0.5 h for infiltration into a hypothetical structured clay soil with equilibration time constants of 0 (local equilibrium), 1, and 2 h.

greater. Note that adding the exponential function in Eq. [4] to describe an increase in conductivity due to macroporosity does not in itself give preferential flow. It simply increases the infiltration rate and gives accelerated advancement of the wetting front. Figure 2 shows that cumulative infiltration was also initially much greater with local equilibrium because of the greater effects of sorption. Lin and McInnes (1995) have noted this effect during infiltration into structured clay soils from disc infiltrometers. Early drainage was also much greater with local equilibrium (Fig. 3), but after 24 h the difference had decreased considerably since the exchange time constants were then relatively less important.

Structured Field Soil

Figure 4 shows the predictions using Richards' equation with local equilibrium of water content changes through time, as compared with the TDR data. The model depicted a distinct wetting front moving progressively through the core and drainage did not commence until 4.5 h, shortly after the wetting front passed the 0.5-m depth (Fig. 4). In contrast, the experimental data exhibited continuing changes to the water content within the core after drainage had commenced much earlier. The improved description of this process using the time constant approach (with fitted time constants for the A and B horizons) is evident in Fig. 5. The time constant model description of the outflow curve for Column 4 is shown in Fig. 6.

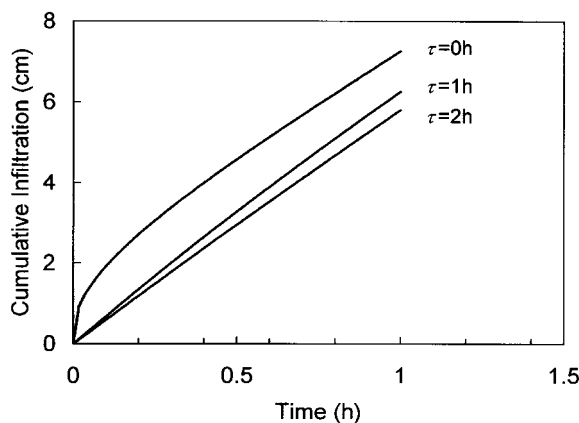


Fig. 2. Cumulative infiltration into a hypothetical structured clay soil with equilibration time constants of 0 (local equilibrium), 1, and 2 h.

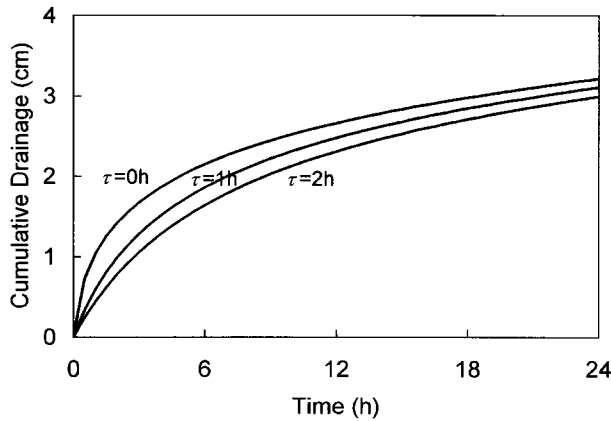


Fig. 3. Cumulative drainage from saturation in a hypothetical structured clay soil with equilibration time constants of 0 (local equilibrium), 1, and 2 h. A unit hydraulic gradient was imposed at the 50-cm depth.

To model Core 3, we altered only K_2 (Eq. [4]) for the surface layer and subsurface clay layer. The value of K_2 for the clay layer was set equal to the final drainage rate of 11 mm h⁻¹ reported by Smettem et al. (1994). The value of K_2 for the surface layer was then adjusted until the best match of the observed drainage and runoff behavior was obtained. At the cessation of irrigation (5 h), the model predicted 46 mm of runoff, compared with 45 mm reported by Smettem et al. (1994). The excellent description of the cumulative outflow is shown in Fig. 6. For a further three of the six cores reported by Smettem et al. (1994), the drainage curves could also be matched by simply varying K_2 for the surface layer and setting K_2 for the clay layer equal to the final drainage rate (data not shown).

Core 1 was traversed by a decaying root and hence exhibited extremely rapid drainage through the base of

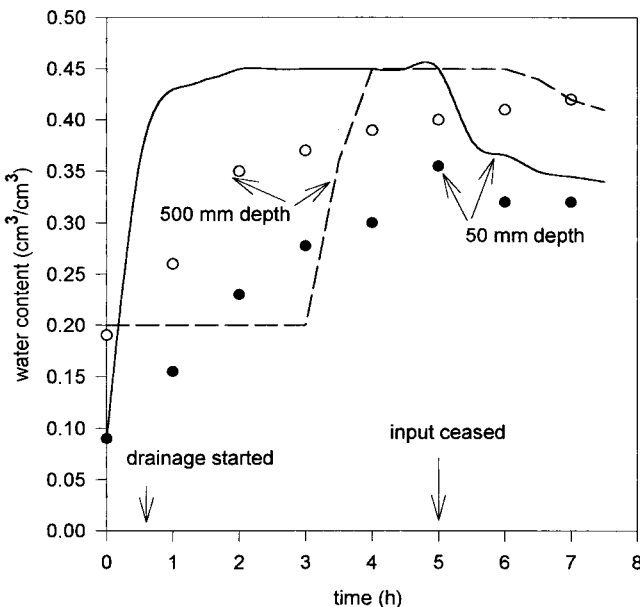


Fig. 4. Water content at two depths during irrigation of Core 4 (Smettem et al., 1994). Observed 50-mm depth (filled circles) and 500-mm depth (open circles). Richards' equation (no time constant) 50-mm depth (solid line) and 500-mm depth (dashed line).

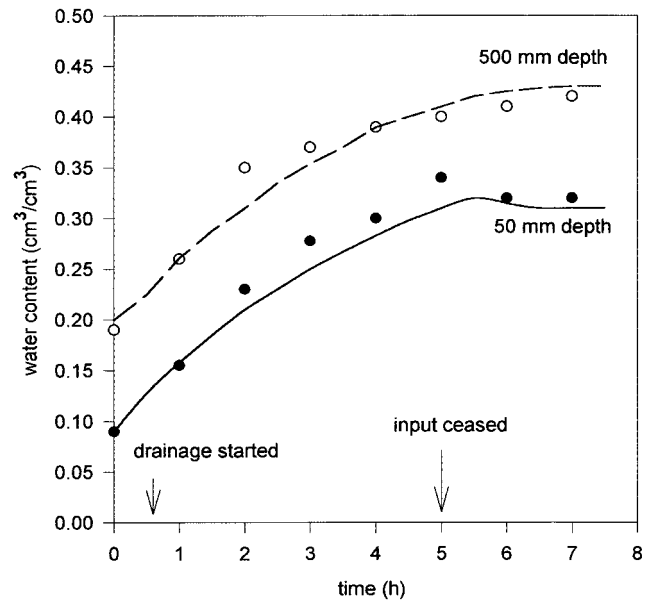


Fig. 5. Water content at two depths during irrigation of Core 4 (Smettem et al., 1994). Observed 50-mm depth (filled circles) and 500-mm depth (open circles). Richards' equation (with time constants from Table 1) 50-mm depth (solid line) and 500-mm depth (dashed line).

the core. To model this behavior, we assumed that K_2 would be high throughout and that the disequilibrium would be extremely large. The curve shown in Fig. 6 was generated by setting K_2 to 2000 mm h⁻¹ for both layers and increasing the time constant τ to 9.2 h, reflecting the time for horizontal movement of water from the root channel. The resulting description of this extreme example of bypass flow was quite reasonable,

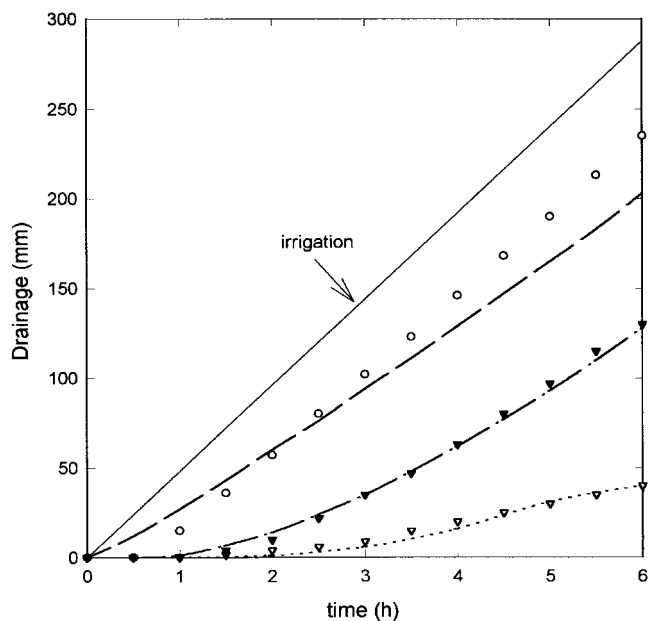


Fig. 6. Cumulative drainage from six undisturbed soil cores under dry antecedent conditions reported by Smettem et al. (1994). The time constant model is shown for three of the cores. Observed drainage: Core 1 (open circles), Core 3 (open triangles), and Core 4 (filled triangles). Time constant model: Core 1 (dashed line), Core 3 (dotted line), and Core 4 (dashed and dotted line).

although application of the one-dimensional Richards equation in this case is questionable.

Estimation of Time Constants

A simple equilibration function can be used to indicate the general effects of local nonequilibrium, but of course a more accurate description depends on a better specification. Philip (1968) discussed nonequilibrium exchange processes in some detail. His investigation suggested that the exact form of the exchange function was not of major importance, within a range of reasonable choices, although his study was directed towards relatively uniform soil aggregates and may not always apply. For example, it is conceivable that some aggregates may have surface properties affecting their absorptive behavior, such as low conductivity skins that would require special consideration. It is likely, however, that for many purposes the simple model proposed here will be adequate, if only because spatial heterogeneity and uncertainty in characterizing the hydraulic properties of field soils limit the accuracy of flow predictions.

The time constant that best describes flow under one set of conditions may not be optimal for another set because the flow geometry may change. This may not prove to be a major disadvantage because, unless equilibration times are very long, with slower flow rates nonequilibrium effects will be less important. The maximum effect will occur with fast moving wetting fronts, and it is under these conditions that time constants need to be estimated. It may be possible to do this by measuring absorption by cores placed in water, provided that these are larger than the structural features causing nonequilibrium. Field measurements using TDR may also prove feasible. These could have the advantage of ensuring that flow conditions were appropriate for the purpose required. For example, during infiltration into a soil with an impeding layer the lower layers do not saturate, so time constants measured on cores would not be applicable for these layers.

There is some hope that a single time constant may characterize a soil horizon, and perhaps may be estimated from a pedological description of the soil when visible soil aggregation is responsible for nonequilibrium. Time constants for Core 4 seemed applicable to Core 3 and to a further three cores for which data has not been presented. The larger values required for Core 1 presumably reflected the root channel present.

CONCLUSIONS

We have presented a simple one-dimensional model of soil water flow that allows effects of nonequilibrium

in matric potential to be described with as little as one additional parameter. Implementation should require only minor changes to most numerical flow codes. We used the model successfully to describe downward flow through cores of a structured field soil. The approach taken is general and macroscopic in the sense that no particular mechanism or geometry is assumed to be responsible for the nonequilibrium. In this it differs from the dual and multiple porosity models. However, similar governing equations can readily be obtained as a special case of a dual porosity model.

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